

Data acquisition + Extraction

INPUT

Literature corpus
Cohort A + Cohort B
934 screened, 50 contributing

Materials Project
Tc + Hc2 + spacegroup
2.2M+ structural records

3DSC canonical cohort
20 fittable on both axes;
52 field-axis fits admitted

Compositional features
Magpie descriptors;
classifier inputs

▼ corpus records: figure, axes, source paper

EXTRACTION

Vision-pass extraction

Round 1/2/3 disambiguation passes under a cross-model agreement gate
Jc(T) at fixed H → β_T | Jc(H) at fixed T → β_H n=662 vision cache entries

Multi-API mainframe

DOI-based retrieval orchestration + UCLA library cross-reference
50 papers over 35 compound labels, 3303 extracted points

▼ curve records: sample form, fixed-axis value

CRITICAL-SCALE RESOLUTION

Assign Tc and Hc2,0 to every fitted curve through the provenance hierarchy
Tier 1 paper-reported · Tier 2 literature-cited · Tier 3 registry · Tier 4 estimated
every curve's Tc and Hc2,0 carries its tier
96 per-paper anchors over 32 papers feed Sec. III.F

methodology,
dispatch, audit

AI TOOLING



Claude (Anthropic)
Methodology + dispatch + audit

Predictor + Empirical Validation + Output

PREDICTOR

Form 3 partial-fit

$\log Jc = \log Jc_{\text{partial}} + \beta \cdot \log_{10} (1 - x/x_c)$
fitted per axis: β_T separately, β_H separately

▼ β_T , β_H and $\log Jc_{\text{partial}}$ per curve

Multi-stage predictor architecture

Stage 1 substructure classification, rank scope
Stage 2 sample-form-conditional median
Stage 3 substructure-aggregate median with an IQR bound
257 temperature-axis fits · 52 field-axis fits from 12 papers

▼ family medians and the regime label

Dispatch through refusal gates

five refusal conditions: reduced field below the validated window,
reduced temperature at or above it, target above Tc,
critical field unavailable, family without a validated field axis

▼ the validated scope, per family and axis

EMPIRICAL VALIDATION

In-corporis

leave-one-compound-out
within substructure
permutation test on the
family label, papers as
the unit
bootstrap CI, N=5000

External

out-of-corpus cuprates
held out entirely
K-anchor sensitivity
at K = 1 and K = 3
measurement-window
saturation reported

▼ envelopes that survive every gate

OUTPUT

Family-scope Jc(T,H) envelope

with a 95% bootstrap interval and an explicit refusal flag
183 candidate compounds evaluated, 84 receive an emitted target

Family-level screening curves

one family curve per grid point; the within-family spread
is a small fraction of the bootstrap interval, and no
per-compound ranking is emitted

Conditioning is a temperature-axis result

the family label accounts for
0.52 of the temperature-exponent
variance between papers,
permutation p = 0.007 (Sec. III.C)

refusal code

the target is withheld

Field axis not validated at family level

the exponent separates no
family on any cohort;
0.038 at p = 0.98 on the
52 admitted fits

183 compounds evaluated

84 receive an emitted target,
all MgB2-class; the other two
families dispatch nothing and
the refusal reason is recorded

resolved Tc and Hc2,0 per curve,
each tagged with its provenance tier